

Marketing Campaign Dataset – Exploratory Data Analysis (EDA) Summary Report

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand campaign performance, clean the data, identify trends, and generate business insights through statistical analysis and visualizations.

Dataset Overview

The dataset contains **200,005 marketing campaign records** with **15 attributes**, including campaign details, marketing channels, customer segments, clicks, impressions, conversion rate, acquisition cost, ROI, engagement score, and campaign dates. The dataset is large enough to perform meaningful analysis and derive business insights.

Data Cleaning

The dataset was examined for quality before analysis. The following preprocessing steps were performed:

- Verified the dataset structure and data types.
- Checked for missing values and found **no missing values** in any column.
- Checked for duplicate records and found **zero duplicate rows**.
- Reviewed descriptive statistics to identify abnormal values.
- Confirmed that the dataset was clean and ready for analysis without requiring additional preprocessing.

Exploratory Data Analysis

Several visualizations were created using **Python, Pandas, Matplotlib, and Seaborn** to explore the dataset. The analysis included distributions of campaign types, marketing channels, customer segments, ROI, conversion rates, acquisition costs, engagement scores, clicks, impressions, and other important business metrics. A total of **11 graphs** were generated to better understand campaign performance and customer behavior.

Marketing Funnel Analysis

The marketing funnel was calculated using the overall campaign metrics:

- **Total Impressions:** 1,101,488,958
- **Total Clicks:** 109,957,667
- **Estimated Conversions:** 8,804,733

The funnel clearly demonstrates the customer journey from impressions to clicks and finally conversions, providing an overview of marketing effectiveness.

Key Business Insights

- The average **ROI** across all marketing channels is approximately **5**, indicating consistent performance among Email, Facebook, Google Ads, Instagram, Website, and YouTube campaigns.
- The average **Conversion Rate** is approximately **8%**, with very little variation across different channels.
- The average **Customer Acquisition Cost** is around **₹12,500**, showing similar spending across marketing platforms.
- **Foodies** represent the largest customer segment, closely followed by **Tech Enthusiasts**, indicating these groups receive the highest campaign focus.
- **Influencer Marketing** is the most frequently used campaign type, followed by **Search, Display, Email**, and **Social Media** campaigns.
- Since performance metrics are relatively balanced across channels, marketers can prioritize channels based on business objectives, target audience, and campaign goals rather than significant differences in ROI.

Conclusion

The dataset was successfully cleaned and analyzed using Python. Exploratory Data Analysis revealed valuable insights into marketing performance, customer segmentation, campaign effectiveness, and acquisition costs. The marketing funnel effectively visualized customer progression from impressions to conversions. Overall, the project demonstrates how data analysis and visualization can support marketing decision-making and optimize future campaign strategies.